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Cantilever with etch chamber flow design

Abstract

A cantilever for gas flow direction control configured to support an electrode housing bowl in an associated etch process chamber. The cantilever may have a cross-section that is circular, elliptical, or airfoil shaped. The shape of the cantilever induces the flow of gas and etch products within the chamber around the cantilever, reducing turbulence around the edge of a wafer.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE (1) This application claims the benefit of U.S. Provisional Application Ser. No. 63/185,572 filed May 7, 2021, and titled NEW CANTILEVER DESIGN WITH ETCH CHAMBER FLOW MECHANICS. U.S. Provisional U.S. Provisional

BACKGROUND

(1) The following relates to systems and components for control of gas flow within an etching chamber. Integrated circuits are formed on a semiconductor substrate, which is typically comprised of silicon. Such formation involves sequential deposition of various materials in layers or films, e.g., conductive and nonconductive layers. Etching processes may be used to form geometric patterns in the layers or vias for electrical contact between the layers. Etching processes may include “wet” etching, wherein a solvent or chemical reagent is used, or “dry” etching, wherein plasma is used. Such processes utilize a gas or fluid for etching, and flow of the gas or fluid impacts wafer quality and uniformity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates an exemplary semiconductor manufacturing apparatus in accordance with some embodiments.

(3) FIG. 2A illustrates a cantilever utilized by the exemplary manufacturing apparatus in accordance with one embodiment of the subject application.

(4) FIG. 2B illustrates a cross-sectional view of the cantilever of FIG. 2A in accordance with one embodiment of the subject application.

(5) FIG. 3A provides an exemplary wafer produced by the semiconductor manufacturing apparatus and cantilever of FIGS. 2A-2B in accordance with one embodiment of the subject application.

(6) FIG. 3B provides an illustrative graph corresponding to the wafer of FIG. 3A.

(7) FIG. 4A provides an illustration of a cylindrical cantilever for use in a semiconductor manufacturing apparatus in accordance with one embodiment of the subject application.

(8) FIG. 4B provides an illustration of the cylindrical cantilever of FIG. 4A during operations of a semiconductor manufacturing apparatus in accordance with one embodiment of the subject application.

(9) FIG. 5A provides an illustration of an elliptical cantilever for use in a semiconductor manufacturing apparatus in accordance with one embodiment of the subject application.

(10) FIG. 5B provides an illustration of the elliptical cantilever of FIG. 5A during operations of a semiconductor manufacturing apparatus in accordance with one embodiment of the subject application.

(11) FIG. 6A provides an illustration of an airfoil/wing cantilever for use in a semiconductor manufacturing apparatus in accordance with one embodiment of the subject application.

(12) FIG. 6B provides an illustration of the airfoil/wing cantilever of FIG. 6A during operations of a semiconductor manufacturing apparatus in accordance with one embodiment of the subject application.

(13) FIG. 7A provides an illustration of an inverted airfoil/wing cantilever for use in a semiconductor manufacturing apparatus in accordance with one embodiment of the subject application.

(14) FIG. 7B provides an illustration of the inverted airfoil/wing cantilever of FIG. 7A during operations of a semiconductor manufacturing apparatus in accordance with one embodiment of the subject application.

(15) FIG. 8 is a flowchart illustrating a method for operating a semiconductor manufacturing apparatus in accordance with one embodiment of the subject application.

DETAILED DESCRIPTION

(16) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(17) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(18) Numerical values in the specification and claims of this application should be understood to include numerical values which are the same when reduced to the same number of significant figures and numerical values which differ from the stated value by less than the experimental error of conventional measurement technique of the type described in the present application to determine the value. All ranges disclosed herein are inclusive of the recited endpoint.

(19) The term “about” can be used to include any numerical value that can vary without changing the basic function of that value. When used with a range, “about” also discloses the range defined by the absolute values of the two endpoints, e.g., “about 2 to about 4” also discloses the range “from 2 to 4.” The term “about” may refer to plus or minus 10% of the indicated number.

(20) Dry etching processes, also referred to as plasma etching processes, are carried out to etch various films at various stages of the semiconductor manufacturing operation and produce various device features. Multiple plasma etching operations are sometimes used during fabrication of a semiconductor device. Various types of plasma etching processes are known in the art, including plasma etching, reactive ion (RI) etching and reactive ion beam etching. In each of these plasma processes, a gas is first introduced into a reaction chamber and then plasma is generated from the gas. This is accomplished by dissociation of the gas into ions, free radicals, and electrons by using an RF (radio frequency) generator, which includes one or more electrodes. The electrodes are accelerated in an electric field generated by the electrodes, and the energized electrons strike gas molecules to form additional ions, free radicals, and electrons, which strike additional gas molecules, and the plasma eventually becomes self-sustaining. The ions, free radicals and electrons in the plasma react chemically with the layer material on the semiconductor wafer to form residual products which leave the wafer surface and thus, etch the material from the wafer.

(21) In the fabrication of semiconductor devices, particularly sub-micron scale semiconductor devices, profiles obtained in the etching process are very important. Careful control of a surface etch process is therefore necessary to ensure directional etching. In conducting an etching process, when an etch rate is considerably higher in one direction than in the other directions, the process is called anisotropic. A reactive ion etching (RIE) process assisted by plasma is frequently used in an anisotropic etching of various material layers on top of S semiconductor substrate. In plasma enhanced etching processes, the etch rate of a semiconductor material is frequently larger than the

sum of the individual etch rates for ion sputtering and individual etching due to a synergy in which chemical etching is enhanced by ion bombardment.

(22) To avoid subjecting a semiconductor wafer to high-energy ion bombardment, the wafer may also be placed downstream from the plasma and outside the discharge area. Downstream plasma etches more in an isotropic manner since there are no ions to induce directional etching. The downstream reactors are frequently used for removing resist or other layers of material where patterning is not critical. In a downstream reactor, radio frequency may be used to generate long-lived radioactive species for transporting to a wafer surface located remote from the plasma. During these etching processes, gas and etch products are introduced and produced, with the flow of such gas and etch products being impacted by the location of the wafer within the chamber, the components, such as a cantilever, supporting the wafer. Such impacted flow can deleteriously affect wafer quality, uniformity, critical depth, and the like.

(23) Turning now to FIG. 1, there is shown a semiconductor manufacturing apparatus **100** in accordance with one embodiment of the subject application. As illustrated in FIG. 1, the semiconductor manufacturing apparatus **100** includes a chamber **102** for accommodating a wafer **104** during operation of the apparatus **100**. An electrode housing bowl **106** is positioned within the chamber **102** and configured to hold the wafer **104** therein. In some embodiments, the electrode housing bowl **106** includes an electrostatic chuck (ESC), a vacuum component, a heater, and/or other mechanism, as well be appreciated by those skilled in the art. In such embodiments, the electrode housing bowl **106** is configured to secure the wafer **104** in a position for processing during operations of the semiconductor manufacturing apparatus **100**.

(24) The electrode housing bowl **106** is depicted in FIG. 1 as being supported within the chamber **102** by a cantilever **114**, extending perpendicularly from an inner wall **116** of the chamber **102** to the center of the chamber. It will be appreciated by those skilled in the art the depiction of the cantilever **114** extending perpendicularly from one inner wall **116** of the chamber **102** is intended as an example only. The cantilever **114** may be attached to any portion of the inner wall **116** of the chamber **102** in accordance with the various embodiments disclosed herein. In some embodiments contemplated herein, in addition to support, power and controls for the electrode housing bowl **106** may be routed through the cantilever **114**.

(25) As shown in FIG. 1, the cantilever **114** supports the electrode housing bowl **106** above a bottom **118** of the chamber **102**. Gas **108**, supplied from an external source, flows into the interior of the chamber **102** via gas inlets **110A**, **110B**, and **110C**. It will be understood that the depicting of three gas inlets **110A-C** is intended as an example, and the semiconductor manufacturing apparatus **100** may include one, two, three, four, five or more inlets to enable gas **108** to enter the chamber **102**. In some embodiments, the gas **108** introduced into the chamber **102** is ignited by a power source such as a radio frequency (RF) generator to generate the plasma **112**. The electrode housing bowl **106** may include an electrode (not shown) configured to ignited and sustain a plasma **112** reaction within the chamber **102**. The plasma **112** may interact with deposition film on the wafer during a dry etch process. The gas **108** may include, for example and without limitation, C_xH_yF_z where x is 1 to 6, y is 0 to 6, and z is 1 to 6; He; Ar; F₂; Cl₂; O₂; N₂; H₂; HBr; HF; NF₃; or SF₆; or the like.

(26) The gas **108**, energized via a coil (not shown) is introduced above the wafer **104** to generate ionized particles. These particles are dropped on the top surface of the wafer **104** and react with the deposited film. Some inert gases, such as nitrogen, argon, or helium may be introduced into the chamber **102** via gas inlets **110A**, **110B**, or **110C** as a carrier gas to distribute the ionized particles more evenly in the chamber **102**. It will be appreciated that inert gas may be used in plasma etching as a diluent and a plasma stabilizer. Diluents may provide a process control variable. For example, an inert gas may be added to increase a total pressure while keeping partial pressures of other gases constant. It may further be appreciated that some gas species may improve the energy transfer from “hot” electrons to reactive gas molecules. Etch products **120**, such as, for example and without

limitation FCN; CO.sub.x where x is 1 or 2; SiCl.sub.x where x is 1 to 4; or SiF.sub.x where x is 1 to 4, may be removed from the chamber **102** via a gas removal mechanism, shown in FIG. **1** as the pump **122**. It will be appreciated that the location of the pump **122** depicted in FIG. **1** is intended as a non-limiting example of the location of the pump **122** relative to the process chamber **102**. The skilled artisan will appreciate that placement of the pump **122** with respect to the chamber **102** may depend upon the type of process chamber, the materials being pumped, the location of the electrode housing bowl **106**, the location of the cantilever **114**, the location of outlets in the chamber **102**, any myriad other factors.

(27) Referring now to FIG. **2A**, there is shown an illustration of a cantilever **202** utilized in the semiconductor manufacturing apparatus **100** in accordance with one embodiment of the subject application. As depicted in FIG. **2A**, the cantilever **200** is illustrated as a uniform support pin, having a square or rectangular cross section **A1 202** and a length **L1 204**. In this example, the square/rectangular cantilever **200** may be positioned such that the top portion/surface of the square/rectangular cantilever cross-section **A1 202** is lower than or equal to the bottom of the wafer **104**, and the bottom portion of the square/rectangular cantilever cross-section **A1 202** is positioned higher than or equal to the top portion of the pump **122**.

(28) FIG. **2B** provides a cross-sectional view of the cantilever **200** during operations of the semiconductor manufacturing apparatus **100**, i.e., as the etch gas **108** is flowing into the chamber **102**. As shown in FIG. **2B**, the configuration of the cantilever **200** may result in turbulence **206** forming on the top side of the cantilever **200**. This is due to the top surface of the cantilever being substantially parallel to the surface of the wafer. The skilled artisan will appreciate that the turbulence **206** may form in the flow of etch gas **108** and/or the flow of etch product **120** during operations of the semiconductor manufacturing apparatus **100**. As will be appreciated by those skilled in the art, the turbulence **206** may impact wafer edge physical quality (e.g., profile, critical dimension, depth uniformity percentage, and the like), polymer defect removal, and the like. Furthermore, it will be appreciated that the foregoing may impact wafer acceptance testing, commercially pure yield enhancements, and the like. That is, high flow turbulence **206** induces problematic wafer-edge physical qualities.

(29) FIG. **3A** provides an illustrative wafer **300** after an etch process performed in the chamber **102** of the semiconductor manufacturing apparatus **100** utilizing the cantilever **200** of FIG. **2A**. As shown in FIG. **3A**, the turbulence **206** formed by the top of the cantilever **200** impacts the wafer edge physical quality, illustrated as the turbulence area **302** in FIG. **3A**. Each diamond shown in FIG. **3A** corresponds to a critical dimension (CD) of the etching pattern on the wafer **300**. It will be appreciated that the diamonds within the turbulence area **302** caused by the cantilever **200** are outside the acceptable SVT CD, with the variance in CD more readily apparent the closer to the edge of the wafer **300**, i.e., variance increases in proximity to the cantilever **200**. The skilled artisan will appreciate that the illustrations presented in FIGS. **3A-3B** are intended solely as non-limiting examples of a wafer **104** produced by the semiconductor manufacturing apparatus **100** of FIG. **1** utilizing the square/rectangular cantilever **200** of FIGS. **2A-2B**.

(30) FIG. **3B** provides a graphical representation **304** of the diamonds on the wafer **300** produced by the semiconductor manufacturing apparatus **100** utilizing the cantilever **200**. The left side of the graph **304** corresponds to the CD referenced above of the etch pattern. The bottom of the graph **304** is divided into four zones (**Z1 306**, **Z2 308**, **Z3 310**, and **Z4 312**), with each zone **Z1-Z4 (306-312)** representative of concentric circles by radius of the wafer **300**. For illustrative purposes, the turbulence area **302**, located in zones **Z3 310** and **Z4 312** more clearly depict the increase in variance of CD values associated with the square/rectangular cross-sectioned cantilever **200** of FIGS. **2A-2B**.

(31) Turning now to FIG. **4A**, there is shown an alternate cantilever **400** for use in the semiconductor manufacturing apparatus **100** in accordance with one embodiment of the subject application. Generally, the cantilever has a curved profile, which reduces turbulence at the edges of

the wafer. As depicted in FIG. 4A, the cantilever **400** is generally cylindrical in shape, having a cross-section **A2 402** and a length **L1 404**. It will be appreciated that the lengths **L1 204** and **404** have the same value in a direct exchange of the cylindrical cantilever **400** for the square/rectangular cantilever **200**, as the length **L1 204, 404** is determined in accordance with the position of the electrode housing bowl **106** within the chamber **102**.

(32) FIG. 4B provides a cross-sectional view of the cylindrical cantilever **400** during operations of the semiconductor manufacturing apparatus **100**, i.e., as gas **108** and/or etch products **120** are flowing within the chamber **102**. In accordance with one exemplary embodiment, the cantilever **400** comprises a cross-sectional area **A2 402** that is defined by: $0.1 \text{ cm} \cdot \text{sup.2} \leq A2 \leq 70685 (\pi \cdot a \cdot b) \text{ cm} \cdot \text{sup.2}$. In such an example, the cantilever **400** utilizes a radius (e.g. **a1 406**, **a2 408**, **b 410**), with the length **A2-1 412** (associated with radius **a1 406**) defined by: $0.1 \text{ cm} \leq b \leq a1 \leq 150 \text{ cm}$ and the length **A2-2 414** (associated with radius **a2 408**) defined by: $0.1 \text{ cm} \leq b \leq a2 \leq 150 \text{ cm}$. The cross-sectional area is considered along the axis joining the electrode housing bowl to the interior wall of the process chamber. In this example, the cylindrical cantilever **400** may be positioned such that the top portion of the cylindrical cantilever cross-section **A2 402** is lower than or equal to the bottom of the wafer **104**, and the bottom portion of the cylindrical cantilever cross-section **A2 402** is positioned higher than or equal to the top portion of the pump **122**.

(33) As will be appreciated by those skilled in the art, the cylindrical cantilever **400** of FIGS. 4A-4B improve the flow mechanics within the chamber **102**. That is, the flow of gas **108** and etch products **120** is improved as they flow around the curved outer surface of the cantilever **400** in contrast to the flat top portion of the cantilever **200** of FIGS. 2A-2B. In accordance with one exemplary embodiment, the flow mechanics provided by the cylindrical (i.e., curved) cantilever **400** prevent or reduce turbulence occurring along the top of the cantilever **400**, in contrast to the cantilever **200** of FIGS. 2A-2B.

(34) In accordance with one example embodiment, the semiconductor manufacturing apparatus **100** of FIG. 1 uses the cylindrical cantilever **400**. In such an example embodiment, the cylindrical cantilever **400** is positioned within the chamber **102**, and affixed at one end to the inner chamber wall **116**. At the opposite end of the cylindrical cantilever **400**, the electrode housing bowl **106** is attached to support a wafer **104** during an etch process.

(35) As an etch gas **108** is introduced into the chamber **102** via the inlets **110A, 1108**, and/or **110C**, an electrode (not shown) within the housing bowl **106** is activated via a suitable source, e.g., an RF match source, high voltage source, or the like. A plasma sheath or cloud **112** is formed over the wafer **104** within the chamber **102** and etching of the wafer **104** occurs. During this etching, the gas **108** is induced to move without turbulence over the cylindrical cantilever **400**, as shown in FIG. 4B. While this is occurring, it will be appreciated that etch products **120** are also induced to move without turbulence over the cylindrical cantilever **400** toward the pump **122** positioned at the bottom of the chamber **102** and configured to extract or remove the extraneous etch gas **108** and/or etch products **120** induced to flow by the cylindrical cantilever **400**. In other embodiments contemplated herein, the pump **122** may be positioned in the center of the bottom **118** of the process chamber **102**, or may be located below the cylindrical cantilever **400**. The skilled artisan will appreciate that the location of the pump **122** may be dependent upon the type of process chamber **102**, the type of etch product **120**, the type of etch gas **108**, and the like.

(36) Referring now to FIG. 5A, there is shown an elliptical cantilever **500** for use in the semiconductor manufacturing apparatus **100** in accordance with another embodiment of the subject application. As shown in FIG. 5A, the cantilever **500** utilizes a generally elliptical component, having a cross section **A3 502** and a length **L1 504**. The skilled artisan will appreciate that the lengths utilized by the cantilevers **200, 400**, and **500** are dependent upon the chamber **102** of the semiconductor manufacturing apparatus **100** in which the cantilever **200, 400, 500** is utilized. That is, depending upon size of the chamber **102**, the length **L1 204, 404**, and/or **504** may change to position the electrode housing bowl **106** accordingly.

(37) FIG. 5B provides a cross-sectional view of the cantilever **500** during operations of the semiconductor manufacturing apparatus **100**. As shown in FIG. 5B, the flow of gas **108** and/or etch products **120** around the elliptical cantilever **500** does not produce turbulence at the top of the elliptical cantilever **500**, as occurring with the square/rectangular shaped cantilever **200** of FIGS. 2A-2B. That is, the flow of gas **108** and/or etch products **120** is not disrupted as they pass around the curve at the apex of the elliptical cross-section. In accordance with one exemplary embodiment, the cross-sectional area $A3$ **502** of the elliptical cantilever **500** is defined by: $0.1 \text{ cm} \cdot \text{sup} \cdot 2 \leq A3 \leq 70685 (\pi \cdot a \cdot b) \text{ cm} \cdot \text{sup} \cdot 2$. In such an example, the elliptical cantilever **500** utilizes a radius (e.g. $a1$ **506**, $a2$ **508**, and width b **510**), with the length $A3-1$ **512** (associated with $a1$ **506**) defined by: $0.1 \text{ cm} \leq b \leq a1 \leq 150 \text{ cm}$ and the length $A3-2$ **514** (associated with $a2$ **508**) defined by: $0.1 \text{ cm} \leq b \leq a2 \leq 150 \text{ cm}$. In this example, the elliptical cantilever **500** may be positioned such that the top portion of the elliptical cantilever cross-section $A3$ **502** is lower than or equal to the bottom of the wafer **104**, and the bottom portion of the elliptical cantilever cross-section $A3$ **502** is positioned higher than or equal to the top portion of the pump **122**.

(38) As will be appreciated by those skilled in the art, the elliptical cantilever **500** of FIGS. 5A-5B improve the flow mechanics within the chamber **102**. The airflow depicted in FIG. 5B demonstrates that the flow of gas **108** and etch products **120** are not obstructed as they flow around the curved outer surface of the elliptical cantilever **500** in contrast to the flat top portion of the cantilever **200** of FIGS. 2A-2B. Accordingly, the flow mechanics provided by the elliptical cantilever **500** prevent or reduce turbulence occurring along the top of the elliptical cantilever **500**.

(39) In accordance with one example embodiment, the semiconductor manufacturing apparatus **100** of FIG. 1 may be modified to utilize the elliptical cantilever **500**. In such an example embodiment, the elliptical cantilever **500** is positioned within the chamber **102**, and affixed at one end to the inner chamber wall **116**. At the opposite end of the elliptical cantilever **500**, the electrode housing bowl **106** is attached to support a wafer **104** during an etch process.

(40) As the etching process begins, a gas **108** is introduced into the chamber **102** via the inlets **110A-110C**, an electrode (not shown) within the housing bowl **106** is activated via a suitable source, e.g., an RF match source, or the like. The gas **108** is then ignited to form a plasma cloud or sheath **112** over the wafer **104** within the chamber **102** and etching of the wafer **104** occurs. During this etching, the gas **108** is induced to move without turbulence over the elliptical cantilever **500**, as shown in FIG. 5B. While this is occurring, it will be appreciated that etch products **120** are also induced to move without turbulence over the elliptical cantilever **500** toward the pump **122** positioned at the bottom of the chamber **102** and configured to extract or remove the extraneous gas **108** and/or etch products **120** induced to flow by the elliptical cantilever **500**.

(41) FIG. 6A depicts a cantilever **600** for use in the semiconductor manufacturing apparatus **100** in accordance with still another embodiment of the subject application. As shown in FIG. 6A, the cantilever **600** utilizes an airfoil/wing shaped component, having a cross section $A4$ **602** and a length $L1$ **604**. As noted above with respect to cantilevers **200**, **400**, and **500**, the lengths $L1$ **204**, $L1$ **404**, $L1$ **504**, and $L1$ **604** are determined in accordance with the chamber **102** of the semiconductor manufacturing apparatus **100** in which the cantilever **200**, **400**, **500**, **600** is utilized. Accordingly, depending upon size of the chamber **102**, the length $L1$ **204**, **404**, **504**, and/or **604** may change to position the electrode housing bowl **106** accordingly.

(42) FIG. 6B provides a cross-sectional view $A4$ **602** of the airfoil/wing cantilever **600** when the semiconductor manufacturing apparatus **100** is processing an associated wafer **104**. As shown in FIG. 6B, the flow of gas **108** and/or etch products **120** around the airfoil/wing cantilever **600** does not produce turbulence at the apex of the airfoil/wing cantilever **600**. That is, the flow of gas **108** and/or etch products **120** is not disrupted as they pass around the leading edge of the airfoil/wing cantilever **600**. In such an example, the airfoil/wing cantilever **600** utilizes a radius (e.g. $a1$ **606**, $a2$ **608**, b **610**). It will be understood that the cross-sectional area of the airfoil/wing cantilever **600** is dependent upon the shape of the airfoil, the upper camber, lower camber, etc., and may include

some flat surfaces. Accordingly, the radius depicted in FIG. 6B is intended as an illustrative example according to one embodiment of the subject application. In accordance with one exemplary embodiment, the airfoil/wing cantilever **600** may be positioned such that the leading edge of the airfoil/wing cantilever **600** is lower than or equal to the bottom of the wafer **104**, and the trailing edge of the airfoil/wing cantilever **600** is positioned higher than or equal to the top portion of the pump **122**.

(43) In some embodiments, the airfoil/wing cantilever **600** may be angled within the chamber **102** to adjust the gas **108** and/or etch product **120** flow toward the pump **122**. That is, in such an embodiment, the pitch of the airfoil/wing cantilever **600** may be determined based upon location of the pump **122** on the bottom of the chamber **102**. For example, if the pump **122** is located towards a front of the chamber **102**, the airfoil/wing cantilever **600** may be angled to direct flow of the gas **108** and/or etch products **120** toward the location of the pump **122**. In other embodiments, the airfoil/wing cantilever **600** is rotatably coupled to the inner wall **116** of the chamber **102**. In such an embodiment, a controller (not shown) may be configured to adjust the pitch of the airfoil/wing cantilever **600** during operations to enable faster or slower flow of gas **108** and/or etch products **120** around the cantilever **600** toward the pump. It will be appreciated by those skilled in the art that the shape of the airfoil/wing cantilever **600** may be adjusted to increase or decrease flow of the gas **108** and/or etch products **120** in accordance with varying types of gas **108** and/or etch products **120** used.

(44) The skilled artisan will appreciate that the airfoil/wing cantilever **600** of FIGS. 6A-6B improve the flow mechanics within the chamber **102**. The airflow depicted in FIG. 6B demonstrates that the flow of gas **108** and etch products **120** are not obstructed as they flow around the curved outer surface of the airfoil/wing cantilever **600** in contrast to the flat top portion of the square/rectangular cantilever **200** of FIGS. 2A-2B. Accordingly, the flow mechanics provided by the airfoil/wing cantilever **600** prevent or reduce turbulence occurring along the leading edge of the airfoil/wing cantilever **600**.

(45) FIG. 7A depicts a cantilever **700** for use in the semiconductor manufacturing apparatus **100** in accordance with still another embodiment of the subject application. As shown in FIG. 7A, the cantilever **700** utilizes an inverted airfoil/wing shaped component, having a cross section A5 **702** and a length L1 **704**. As noted above with respect to cantilevers **200**, **400**, **500**, and **600** the lengths L1 **204**, L1 **404**, L1 **504**, L1 **604**, and L1 **704** are determined in accordance with the chamber **102** of the semiconductor manufacturing apparatus **100** in which the cantilever **200**, **400**, **500**, **600**, **700** is utilized. Accordingly, depending upon size of the chamber **102**, the length L1 **204**, **404**, **504**, **604**, and/or **704** may change to position the electrode housing bowl **106** accordingly.

(46) FIG. 7B provides a cross-sectional view A5 **702** of the inverted airfoil/wing cantilever **700** when the semiconductor manufacturing apparatus **100** is processing an associated wafer **104**. As shown in FIG. 7B, the flow of gas **108** and/or etch products **120** around the inverted airfoil/wing cantilever **700** does not produce turbulence at the apex of the inverted airfoil/wing cantilever **700**. That is, the flow of gas **108** and/or etch products **120** is not disrupted as they pass around the trailing edge of the inverted airfoil/wing cantilever **700**. In such an example, the inverted airfoil/wing cantilever **700** utilizes a radius (e.g. a1 **706**, a2 **708**, b **710**). It will be understood that the cross-sectional area of the airfoil/wing cantilever **700** is dependent upon the shape of the airfoil, the upper camber, lower camber, etc., and may include some flat surfaces. Accordingly, the radius depicted in FIG. 7B is intended as an illustrative example according to one embodiment of the subject application. In accordance with one exemplary embodiment, the inverted airfoil/wing cantilever **700** may be positioned such that the trailing edge of the inverted airfoil/wing cantilever **700** is lower than or equal to the bottom of the wafer **104**, and the leading edge of the inverted airfoil/wing cantilever **700** is positioned higher than or equal to the top portion of the pump **122**.

(47) In some embodiments, the inverted airfoil/wing cantilever **700** may be angled within the chamber **102** to adjust the gas **108** and/or etch product **120** flow toward the pump **122**. That is, in

such an embodiment, the pitch of the inverted airfoil/wing cantilever **700** may be determined based upon location of the pump **122** on the bottom of the chamber **102**. For example, if the pump **122** is located towards a front of the chamber **102**, the inverted airfoil/wing cantilever **700** may be angled to direct flow of the gas **108** and/or etch products **120** toward the location of the pump **122**. In other embodiments, the inverted airfoil/wing cantilever **700** is rotatably coupled to the inner wall **116** of the chamber **102**. In such an embodiment, a controller (not shown) may be configured to adjust the pitch of the inverted airfoil/wing cantilever **700** during operations to enable faster or slower flow of gas **108** and/or etch products **120** around the cantilever **700** toward the pump. It will be appreciated by those skilled in the art that the shape of the inverted airfoil/wing cantilever **700** may be adjusted to increase or decrease flow of the gas **108** and/or etch products **120** in accordance with varying types of gas **108** and/or etch products **120** used.

(48) The skilled artisan will appreciate that the inverted airfoil/wing cantilever **700** of FIGS. 7A-7B improves the flow mechanics within the chamber **102**. The airflow depicted in FIG. 7B demonstrates that the flow of gas **108** and etch products **120** are not obstructed as they flow around the curved outer surface of the inverted airfoil/wing cantilever **700** in contrast to the flat top portion of the square/rectangular cantilever **200** of FIGS. 2A-2B. Accordingly, the flow mechanics provided by the inverted airfoil/wing cantilever **700** prevent or reduce turbulence occurring along the trailing edge of the inverted airfoil/wing cantilever **700**.

(49) In accordance with one example embodiment, the semiconductor manufacturing apparatus **100** of FIG. 1 may be modified to utilize the airfoil/wing cantilever **600** described above in place of the cantilever **114**. In such an example embodiment, the airfoil/wing cantilever **600** is positioned within the chamber **102**, and rotatably attached at one end to the inner chamber wall **116**. At the opposite end of the airfoil/wing cantilever **600**, the electrode housing bowl **106** is attached to support a wafer **104** during an etch process. The attachments at the inner wall **116** and the electrode housing bowl **106** may be movable to enable the airfoil/wing cantilever **600** to rotate during operations of the etching process. In some embodiments contemplated herein, a motor is affixed to the inner wall **116** and/or the housing **106** to rotate the airfoil/wing cantilever **600**. The motor may be controlled by an external source, such as a controller. Alternatively, the airfoil/wing cantilever **600** may be rotatable solely via the flow of the gas **108** and/or etch product **120** over the surface thereof. That is, the direction of the flow of gas **108** and/or etch product **120** over the airfoil/wing cantilever **600** may cause the airfoil/wing cantilever **600** to rotate about its axis in response to the direction of the gas **108** and/or etch product **120** flow. The skilled artisan will appreciate that such movement may improve flow within the chamber **102**, thereby increasing the quality of the wafer **104** as inlet gas **108** is directed over the wafer **104** and etch product **120** is directed away from the wafer **104**.

(50) As a gas **108** is introduced into the chamber **102** via the inlets **110A-C**, an electrode (not shown) located within the housing bowl **106** is activated via a suitable source, e.g., an RF match source, high voltage generator, or the like. A plasma sheath or cloud **112** is formed over the wafer **104** within the chamber **102** and etching of the wafer **104** occurs. During this etching, the gas **108** is induced to move without turbulence over the airfoil/wing cantilever **600**, as shown in FIG. 6B. While this is occurring, it will be appreciated that etch products **120** are formed from removal of materials from the wafer **104** and are also induced to move without turbulence over the airfoil/wing cantilever **600** toward the pump **122** positioned at the bottom of the chamber **102** and configured to extract or remove the extraneous gas **108** and/or etch products **120** induced to flow by the airfoil/wing cantilever **600**.

(51) It will be appreciated by those skilled in the art that the use of a dry etch process is intended solely to be one example implementation of the foregoing cantilever **400**, **500**, **600** embodiments. The designs set forth above are configured to induce the flow of any suitable fluid within a wet, dry, or other semiconductor wafer processing chamber. As such, the skilled artisan will appreciate that the cantilever **400**, **500**, **600** may be utilized to support a wafer **104** for processing in a myriad of varying process chambers.

(52) FIG. 8 is a flowchart illustrating a method for operating a semiconductor manufacturing apparatus in accordance with one embodiment of the subject application. In step **810**, the wafer is positioned on an electrode housing bowl within an etch process chamber. In step **820**, an etch gas is flowed past the wafer to etch the wafer. In step **830**, an etch product is removed from the etch process chamber, for example using a pump. It is noted that steps **820** and **830** can be considered as occurring concurrently. In step **840**, the etch gas flow is stopped. The pumping may continue until the environment of the etch process chamber reaches a safe state. After a suitable time period, the etched wafer can be removed from the etch process chamber.

(53) In accordance with one embodiment, there is provided a cantilever for gas flow direction control configured to support an electrode housing bowl in an associated etch process chamber. The cantilever is coupled to an inner wall of the associated process chamber. The cantilever has a curved profile, for example a cylindrical, elliptical, or airfoil/wing shape.

(54) In accordance with another embodiment, there is provided an etch process chamber that includes an electrode housing bowl positioned in a center of the interior of the process chamber. A cantilever couples the electrode housing bowl to an interior wall of the process chamber. The cantilever has a curved profile to reduce turbulence. In addition, the etch process chamber comprises a pump configured to remove at least one of an etch gas or an etch product from the interior of the process chamber.

(55) In accordance with still another embodiment, there is provided a semiconductor manufacturing apparatus. The semiconductor manufacturing apparatus includes an etch process chamber that comprises a cantilever and an electrode housing bowl. The cantilever is coupled to an interior wall of the process chamber and is configured to direct a flow of at least one of an etch gas or an etch product within the process chamber. The cantilever has a cross-sectional profile that is cylindrical, elliptical, or an airfoil. The electrode housing bowl is positioned in the center of the interior of the process chamber and coupled to the cantilever. The semiconductor manufacturing apparatus also comprises a pump positioned outside the etch process chamber and configured to remove at least one of an etch gas or an etch product from the interior of the process chamber.

(56) In accordance with another embodiment, there is provided a method of etching a wafer. The method includes positioning the wafer on an electrode housing bowl within an etch process chamber. The method also includes flowing an etch gas past the wafer to etch the wafer. In addition, the method includes removing an etch product from the etch process chamber. In such an embodiment, the electrode housing bowl is coupled to an interior wall of the process chamber by a cantilever, the cantilever having a curved profile to reduce turbulence.

(57) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An etch process chamber, comprising: an electrode housing bowl positioned in a center of an interior of the etch process chamber, the etch process chamber including a first portion where an etch gas enters the interior of the etch process chamber and a second portion where the etch gas exits the interior of the etch process chamber, and the electrode housing bowl configured to hold a wafer with a wafer surface oriented towards the first portion; a gas inlet positioned within the etch process chamber first portion; a pump configured to remove at least one of the etch gas or an etch

product from the interior of the process chamber via the second portion of the etch process chamber; and a cantilever coupling the electrode housing bowl to an interior wall of the etch process chamber, the cantilever configured to route one or both of a power and a control cable through the cantilever coupling to the electrode housing bowl, the cantilever having an airfoil shape, and the cantilever having a continuous curved outer surface profile to reduce turbulence of the etch gas as it flows through the interior of the etch process chamber from the etch gas inlet around the cantilever coupling and towards the pump and out of the etch process chamber, and the airfoil shape having a leading edge positioned against a flow direction of the etch gas as it flows through the interior of the etch process chamber from the etch gas inlet; wherein the cantilever airfoil shape leading edge has a leading edge height lower than or equal to a height of the wafer, the cantilever airfoil shape has a trailing edge having a trailing edge height higher than or equal to a height of the pump, the leading edge has a surface profile with a first radius, and the trailing edge has a surface profile with a second radius less than the first radius to provide an apex of the cantilever airfoil shape.

2. The etch process chamber of claim 1, wherein the cantilever is located to induce a flow of at least one of the etch gas or the etch product to an associated wafer positioned on the electrode housing bowl.

3. The etch process chamber of claim 1, wherein the cantilever is located to induce a flow of at least one of the etch gas or the etch product to the pump.

4. The etch process chamber of claim 1, wherein the etch gas comprises $C_xH_yF_z$ where x is 1 to 6, y is 0 to 6, and z is 1 to 6; He; Ar; F_2 ; Cl_2 ; O_2 ; N_2 ; H_2 ; HBr; HF; NF_3 ; or SF_6 .

5. The etch process chamber of claim 1, wherein the etch product comprises FCN; CO_x where x is 1 or 2; $SiCl_x$ where x is 1 to 4; or SiF_x where x is 1 to 4.

6. A semiconductor manufacturing apparatus, comprising: an etch process chamber, comprising: a cantilever coupled to an interior wall of the process chamber, the cantilever configured to direct a flow of at least one of an etch gas or an etch product within the etch process chamber, an electrode housing bowl positioned in a center of the interior of the etch process chamber and coupled to the cantilever; and a pump positioned outside the etch process chamber and configured to remove at least one of an etch gas or an etch product from the interior of the etch process chamber; wherein the cantilever is configured to route one or both of a power and a control cable through the cantilever to the electrode housing bowl, the cantilever has an airfoil shape and the cantilever has a continuous curved outer surface profile to reduce turbulence of the etch gas as it flows through the interior of the etch process chamber from an etch gas inlet around the cantilever and towards the pump and out of the etch process chamber, and the airfoil shape has a leading edge positioned against a flow direction of the etch gas as it flows through the interior of the etch process chamber from the etch gas inlet; wherein the cantilever airfoil shape has a leading edge portion extending along a first length of the cantilever, a trailing edge portion extending along a second length of the cantilever, and a maximum width of the leading edge portion is different from a maximum width of the trailing edge portion.

7. The semiconductor manufacturing apparatus of claim 6, wherein the cantilever is located to induce a flow of at least one of the etch gas or the etch product to an associated wafer positioned on the electrode housing bowl.

8. The semiconductor manufacturing apparatus of claim 6, wherein the cantilever is located to induce a flow of at least one of the etch gas or the etch product to the pump.

9. The semiconductor manufacturing apparatus according to claim 6, wherein the maximum width of the leading edge portion is greater than the maximum width of the trailing edge portion.

10. The semiconductor manufacturing apparatus according to claim 6, wherein the maximum width of the leading edge portion is less than the maximum width of the trailing edge portion.

11. The semiconductor manufacturing apparatus of claim 6, wherein the etch gas comprises

C.sub.xH.sub.yF.sub.z where x is 1 to 6, y is 0 to 6, and z is 1 to 6; He; Ar; F.sub.2; Cl.sub.2; O.sub.2; N.sub.2; H.sub.2; HBr; HF; NF.sub.3; or SF.sub.6.

12. The semiconductor manufacturing apparatus of claim 6, wherein the etch product comprises FCN; CO.sub.x where x is 1 or 2; SiCl.sub.x where x is 1 to 4; or SiF.sub.x where x is 1 to 4.

13. A semiconductor wafer etch process chamber, comprising: an electrode housing bowl positioned in a center of an interior of the etch process chamber, the etch process chamber including a first portion where an etch gas enters an interior of the etch process chamber and a second portion where the etch gas exits the interior of the etch process chamber, and the electrode housing bowl configured to hold a wafer with a wafer surface oriented towards the first portion; a gas inlet positioned at the etch process chamber first portion; a pump configured to remove at least one of the etch gas or an etch product from the interior of the etch process chamber via the second portion of the etch process chamber; and a cantilever coupling the electrode housing bowl to an interior wall of the etch process chamber, the cantilever configured to route one or both of a power and a control cable through the cantilever to the electrode housing bowl, the cantilever having an inverted airfoil shape and the cantilever having a continuous curved outer surface profile to reduce turbulence of the etch gas as it flows through the interior of the etch process chamber from the etch gas inlet around the cantilever and towards the pump and out of the etch process chamber, and the inverted airfoil shape having a leading edge positioned against a flow direction of the etch gas as it flows through the interior of the etch process chamber from the etch gas inlet; wherein the cantilever inverted airfoil shape leading edge has a leading edge height lower than or equal to a bottom height of the wafer, the cantilever inverted airfoil shape has a trailing edge having a trailing edge height higher than or equal to a top portion height of the pump, the leading edge has a surface profile with a first radius, and the trailing edge has a surface profile with a second radius greater than the first radius where the first radius provides an apex of the cantilever inverted airfoil shape.

14. The semiconductor wafer etch process chamber according to claim 13, wherein the cantilever is located to induce a flow of at least one of the etch gas or the etch product to a wafer secured to the electrode housing bowl.

15. The semiconductor wafer etch process chamber of claim 13, wherein the etch gas comprises C.sub.xH.sub.yF.sub.z where x is 1 to 6, y is 0 to 6, and z is 1 to 6; He; Ar; F.sub.2; Cl.sub.2; O.sub.2; N.sub.2; H.sub.2; HBr; HF; NF.sub.3; or SF.sub.6.

16. The semiconductor wafer etch process chamber of claim 13, wherein the etch product comprises FCN; CO.sub.x where x is 1 or 2; SiCl.sub.x where x is 1 to 4; or SiF.sub.x where x is 1 to 4.
